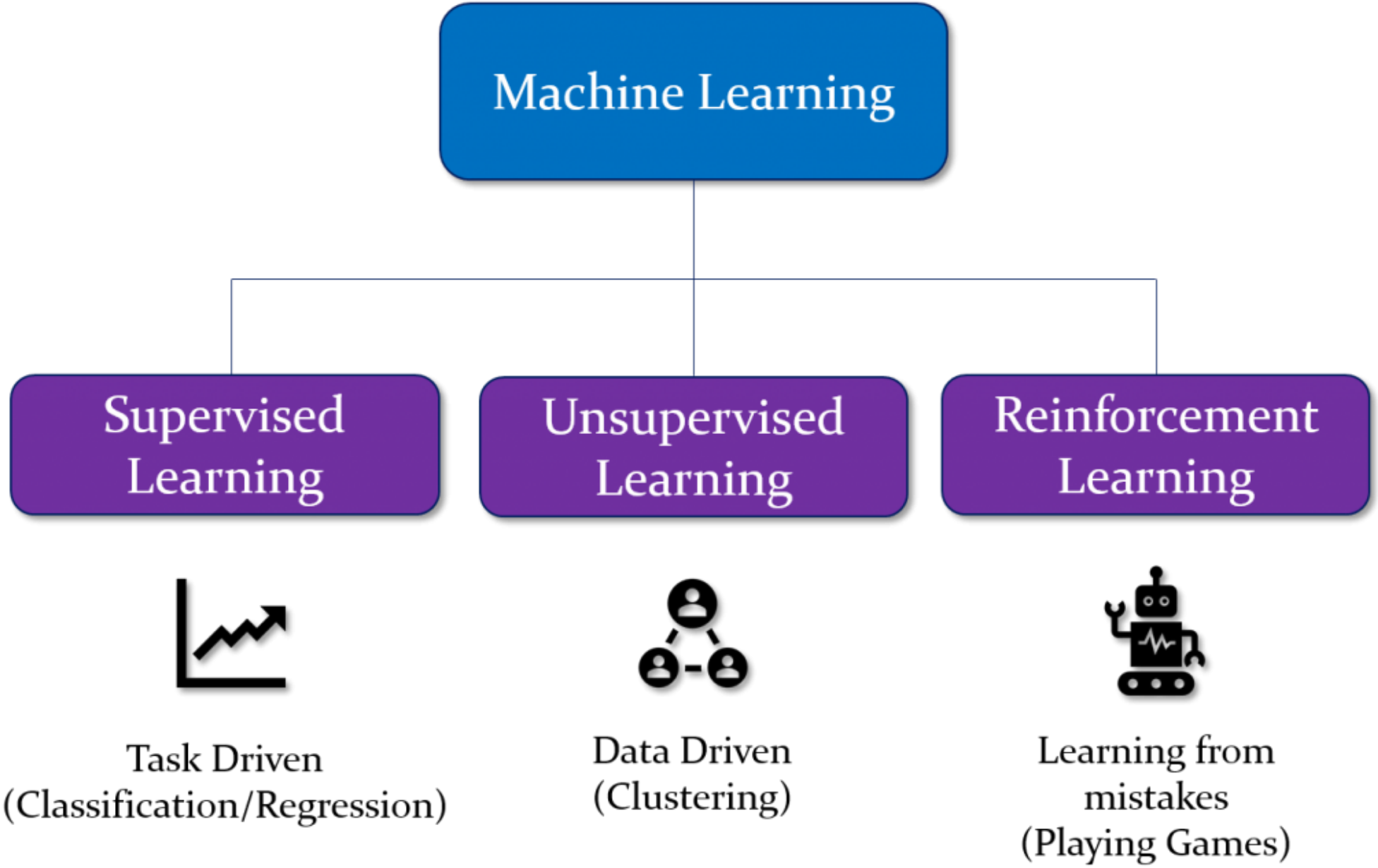
# Unsupervised Machine Learning

Clustering

# Overview

- Types of machine learning
- Unsupervised learning
  - K Means Clustering
    - The intuition
    - A visual demo
    - A practical demo
  - Hierarchical Clustering

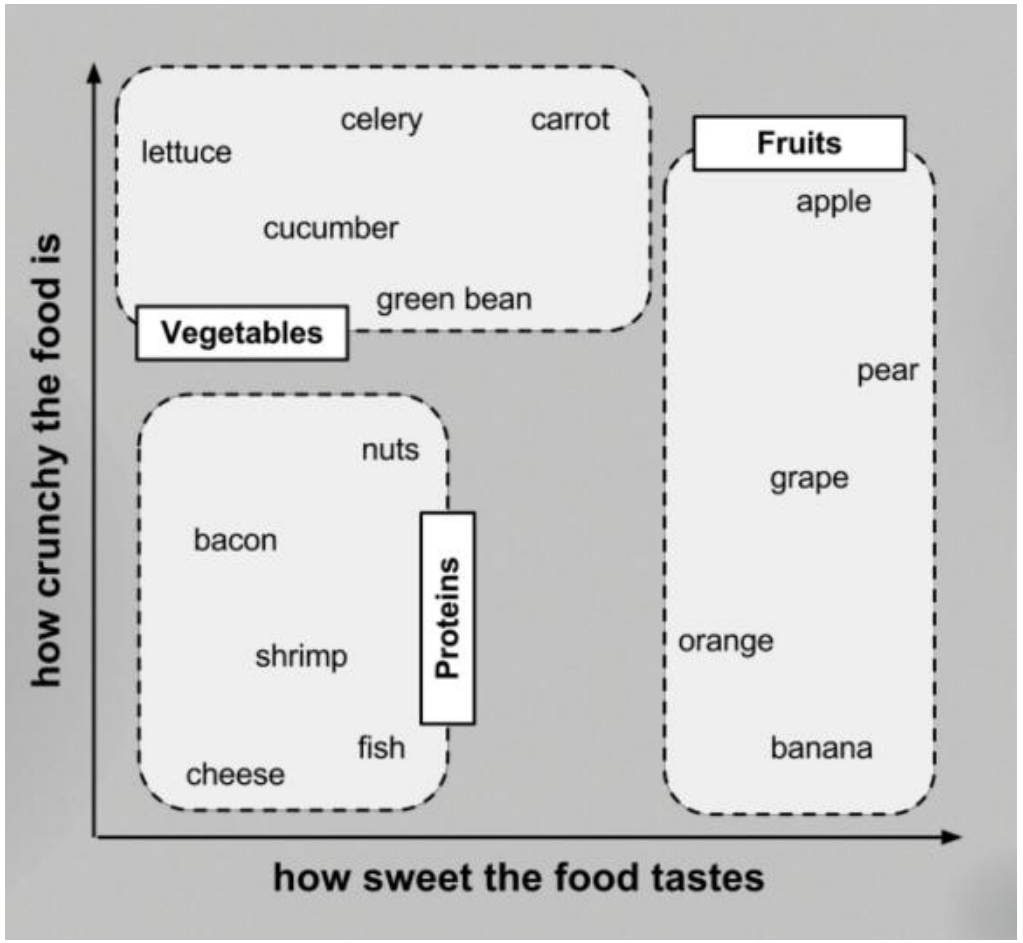
# Types of Machine Learning



# Supervised vs Unsupervised Learning

- Example from kNN

| ingredient | sweetness | crunchiness | food type |
|---|---|---|---|
| apple | 10 | 9 | fruit |
| bacon | 1 | 4 | protein |
| banana | 10 | 1 | fruit |
| carrot | 7 | 10 | vegetable |
| celery | 3 | 10 | vegetable |
| cheese | 1 | 1 | protein |



# Supervised vs Unsupervised Learning

| ingredient | sweetness | crunchiness |
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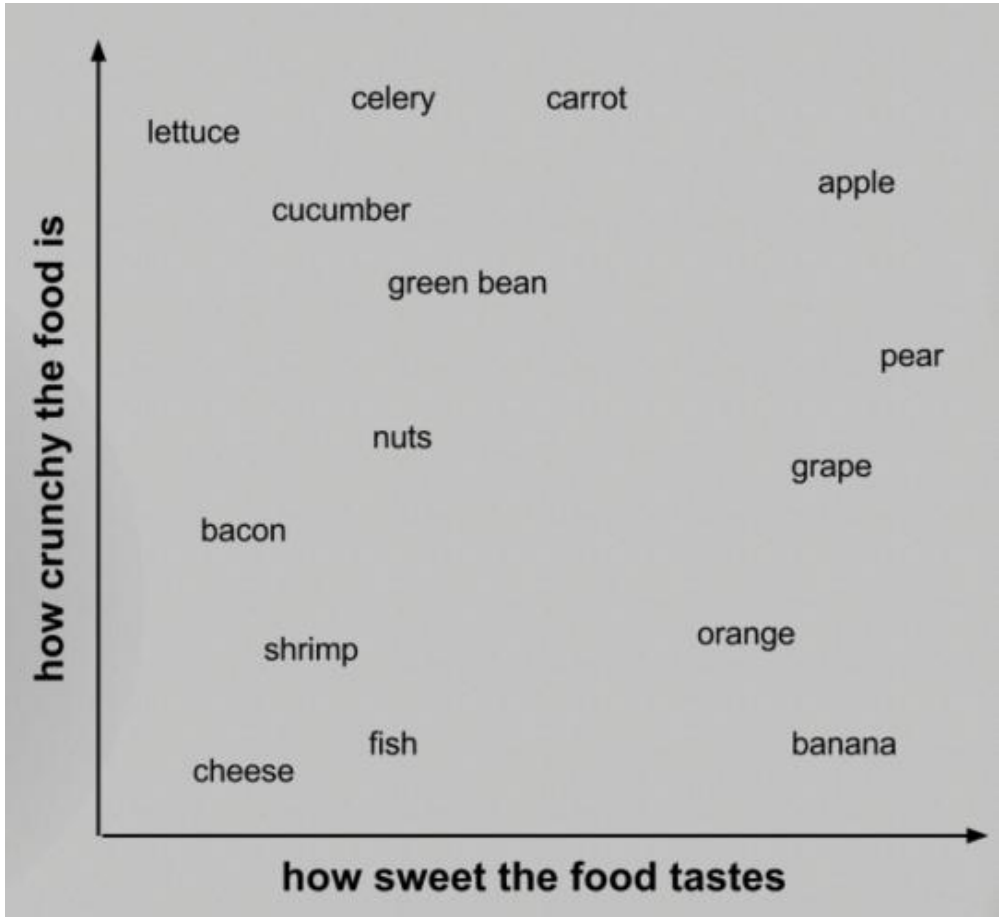

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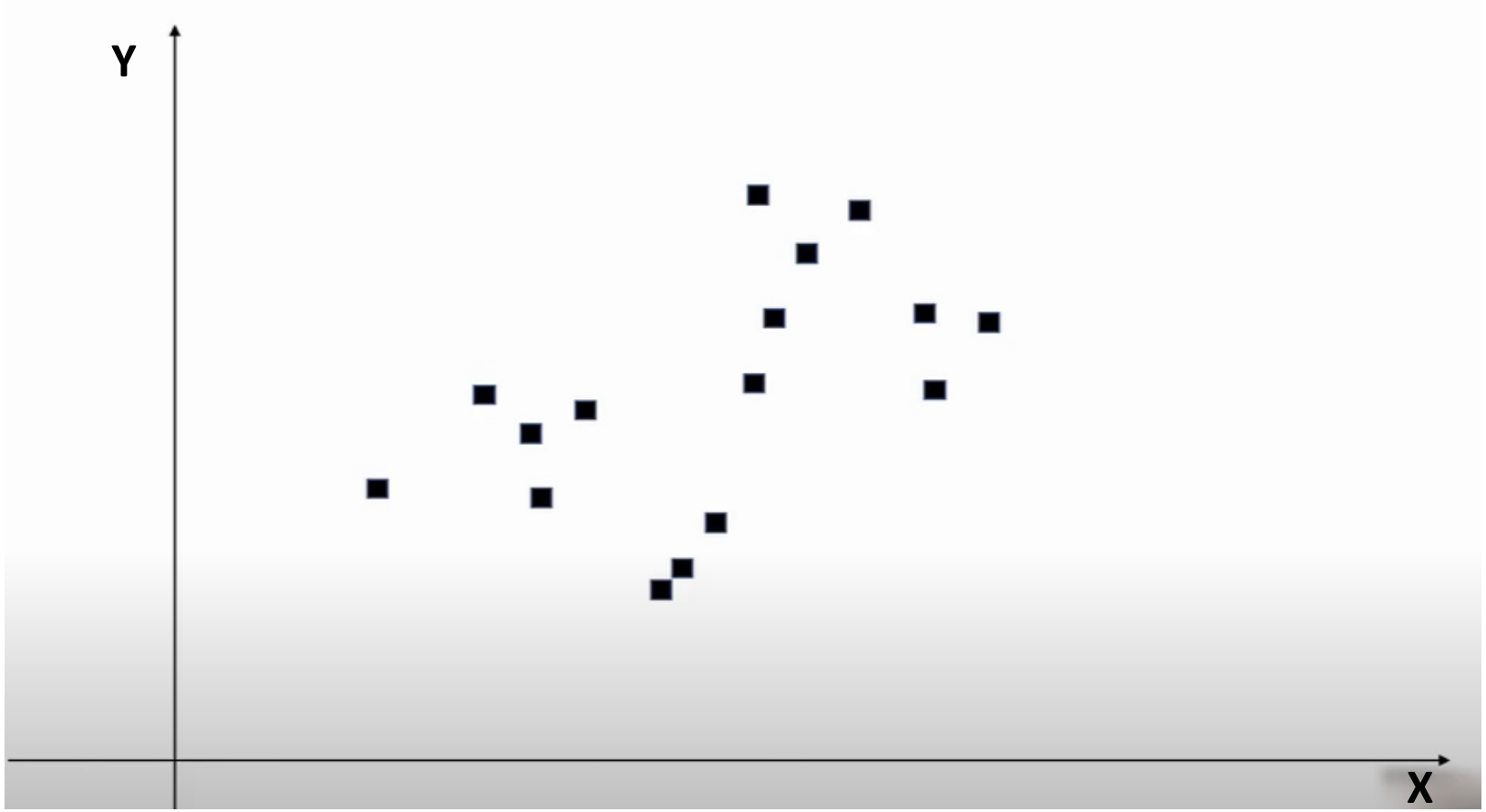
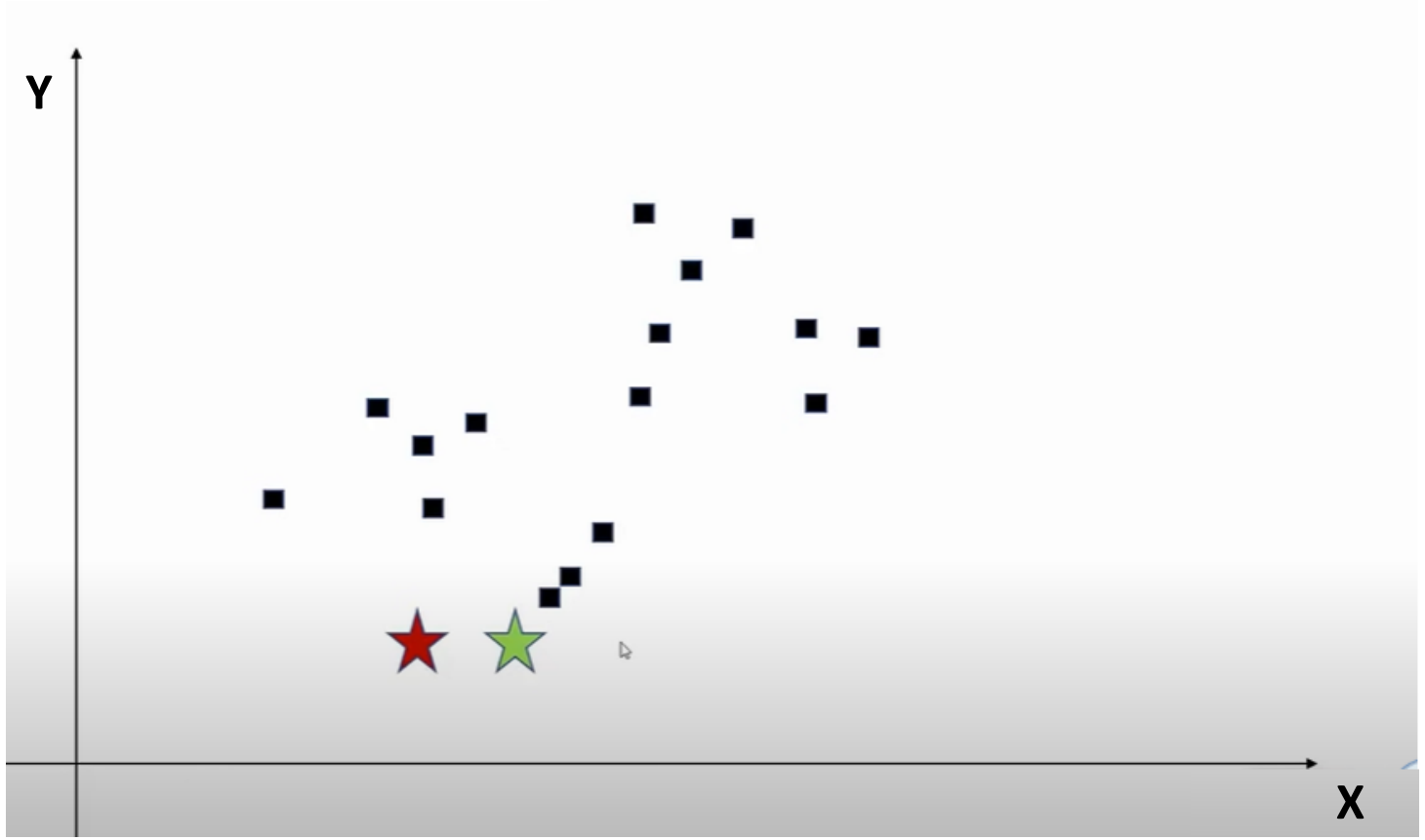
# K Means Clustering (The Intuition)



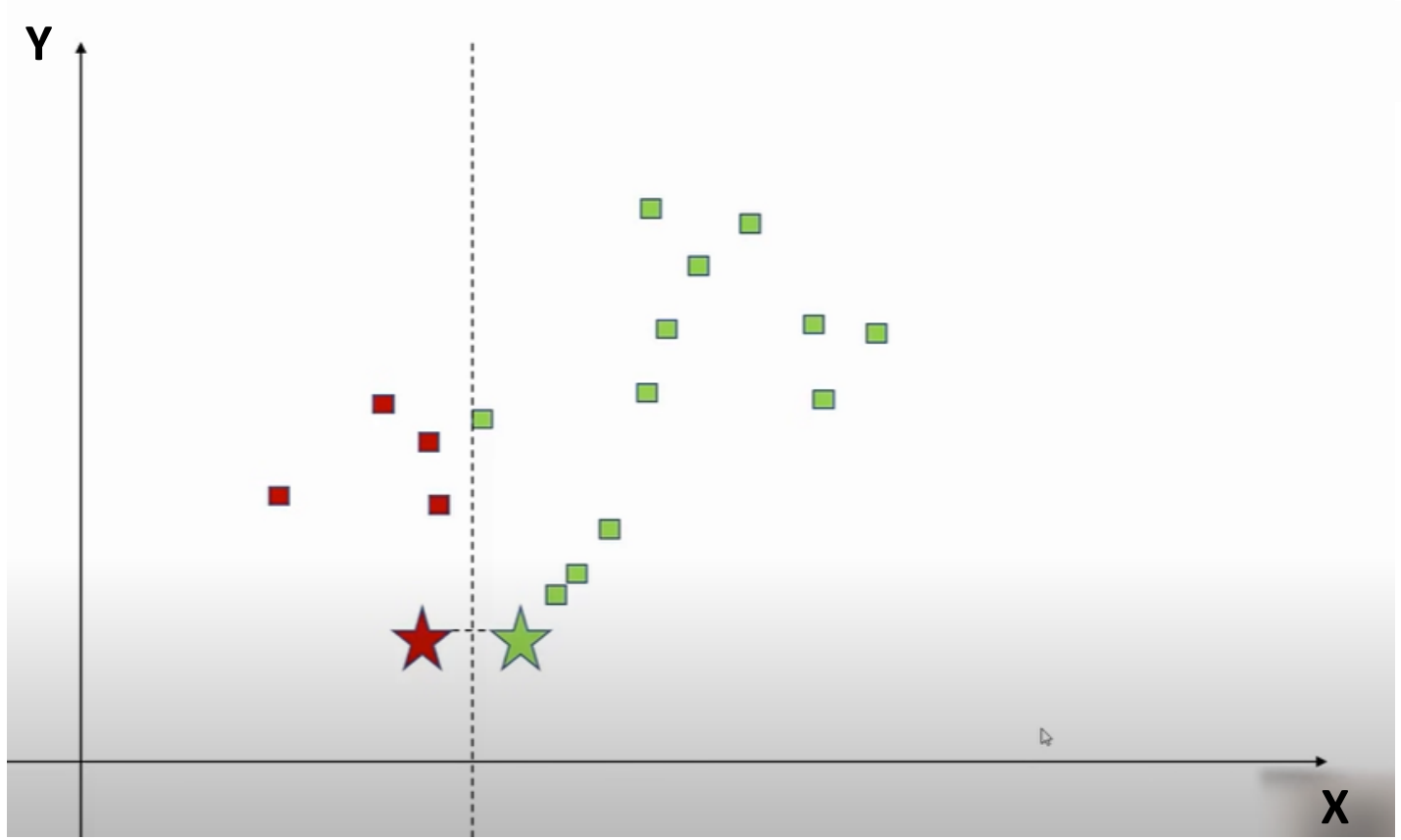
# K Means Clustering (The Intuition)

- First step: Start with K centroids by putting them in random place
  - Here k = 2



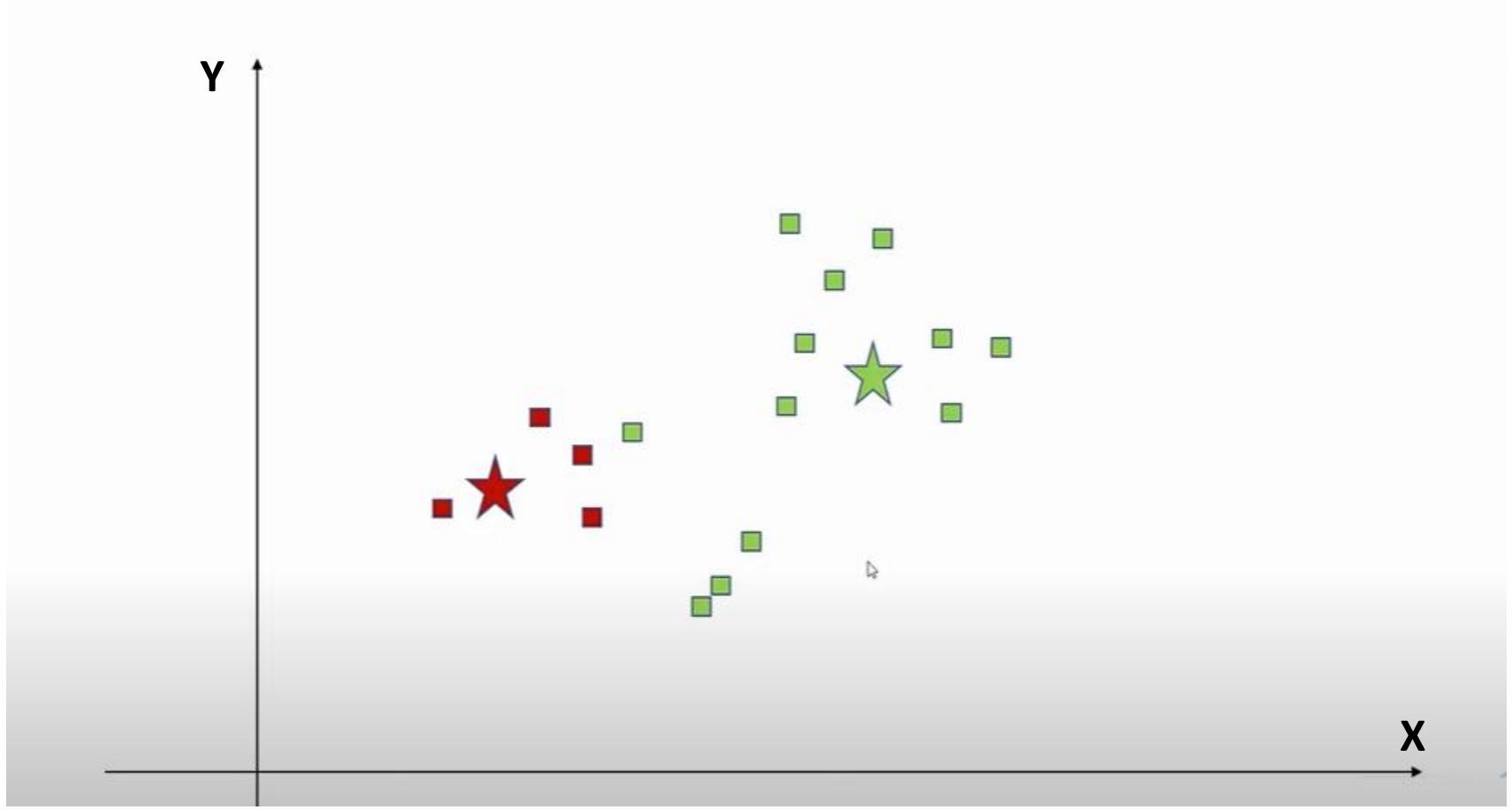
# K Means Clustering (The Intuition)

- Second step: Compute distance of every point from centroid and cluster them accordingly



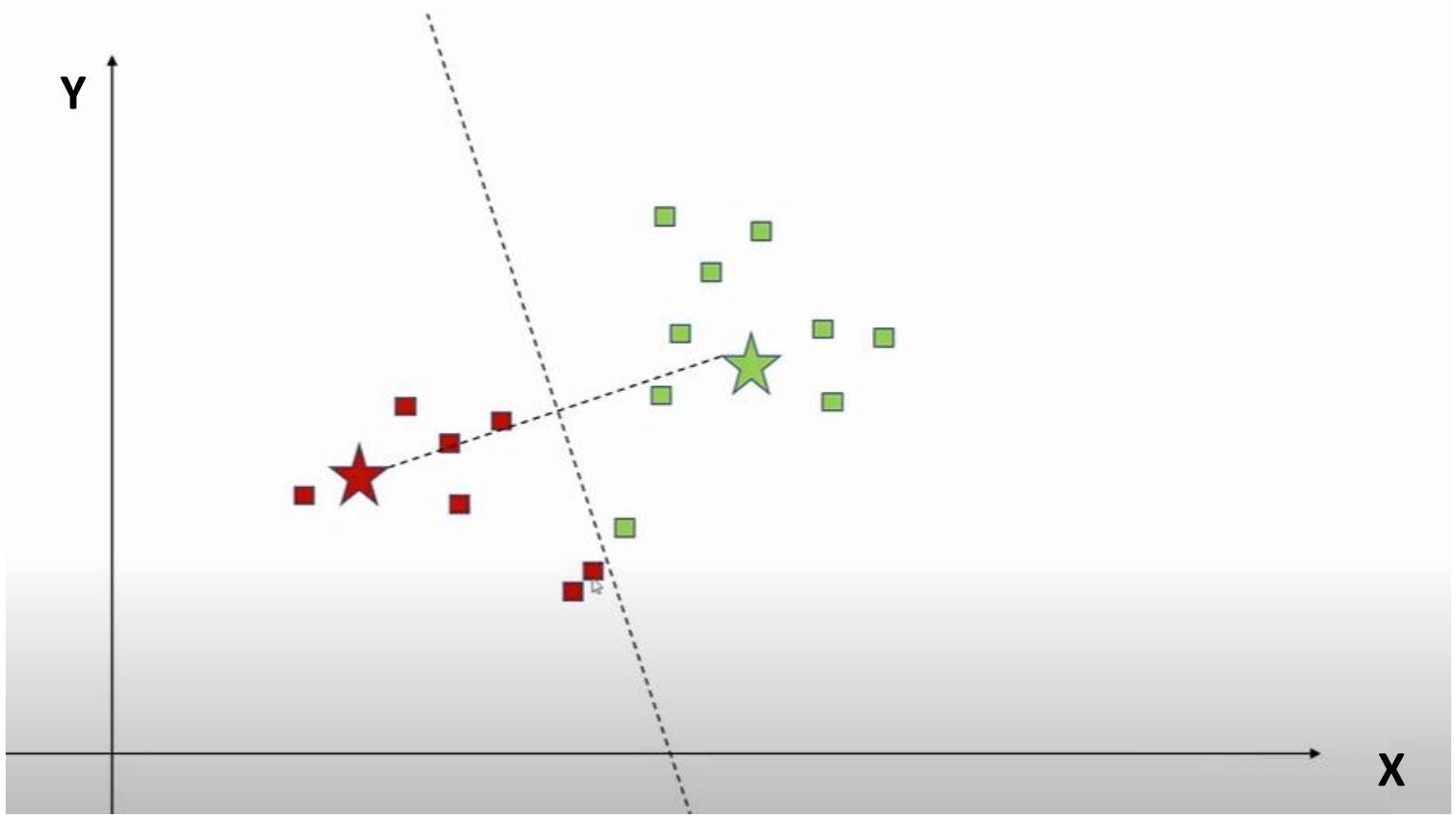
# K Means Clustering (The Intuition)

- Third step: Adjust centroid so that they become center of gravity for given cluster



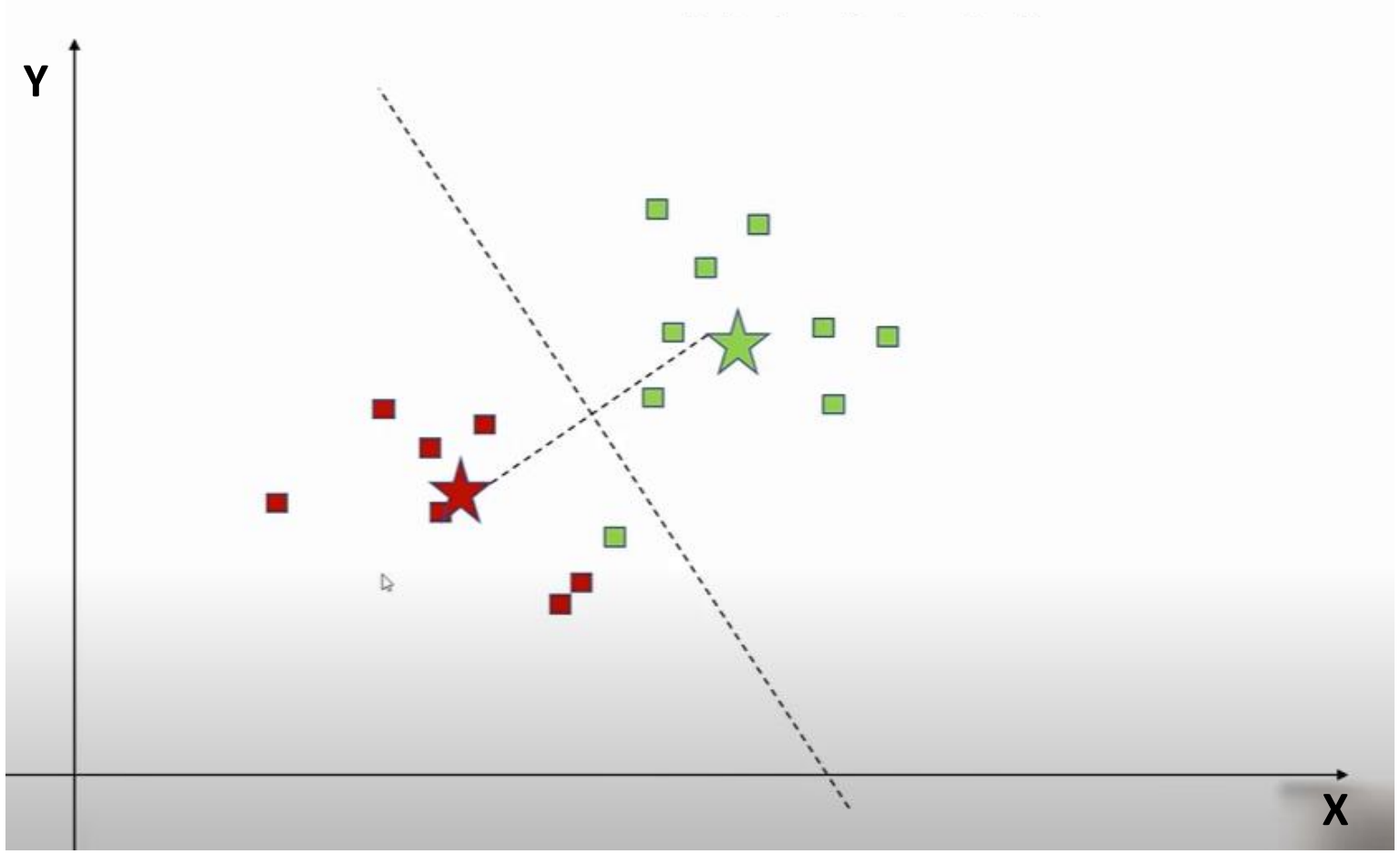
# K Means Clustering (The Intuition)

- Fourth step: Again, re-cluster every point based on their distance with centroid



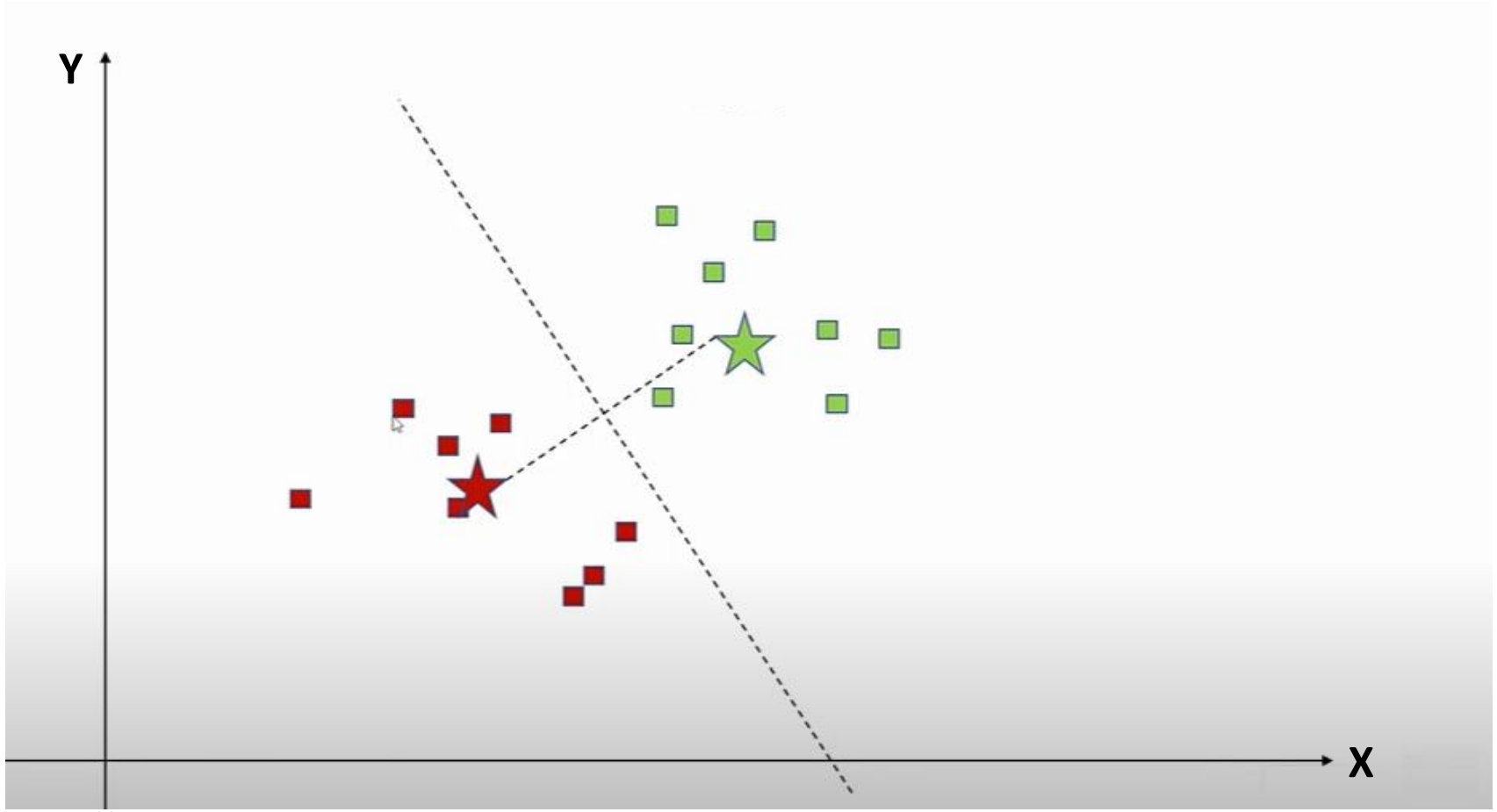
# K Means Clustering (The Intuition)

- Fifth step: Again, Adjust centroids

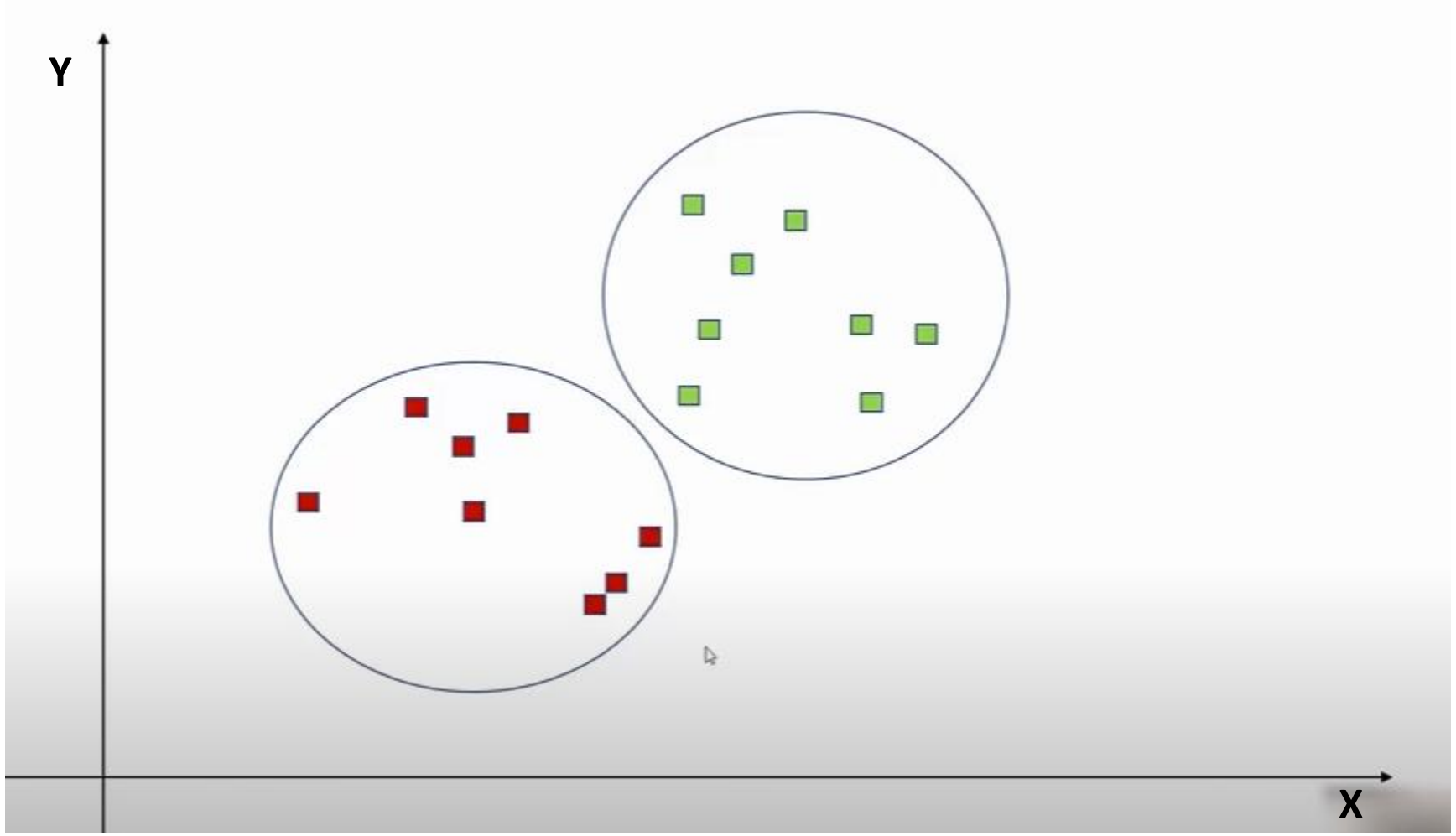


# K Means Clustering (The Intuition)

- Sixth step: Recompute clusters and repeat this till data points stop changing clusters



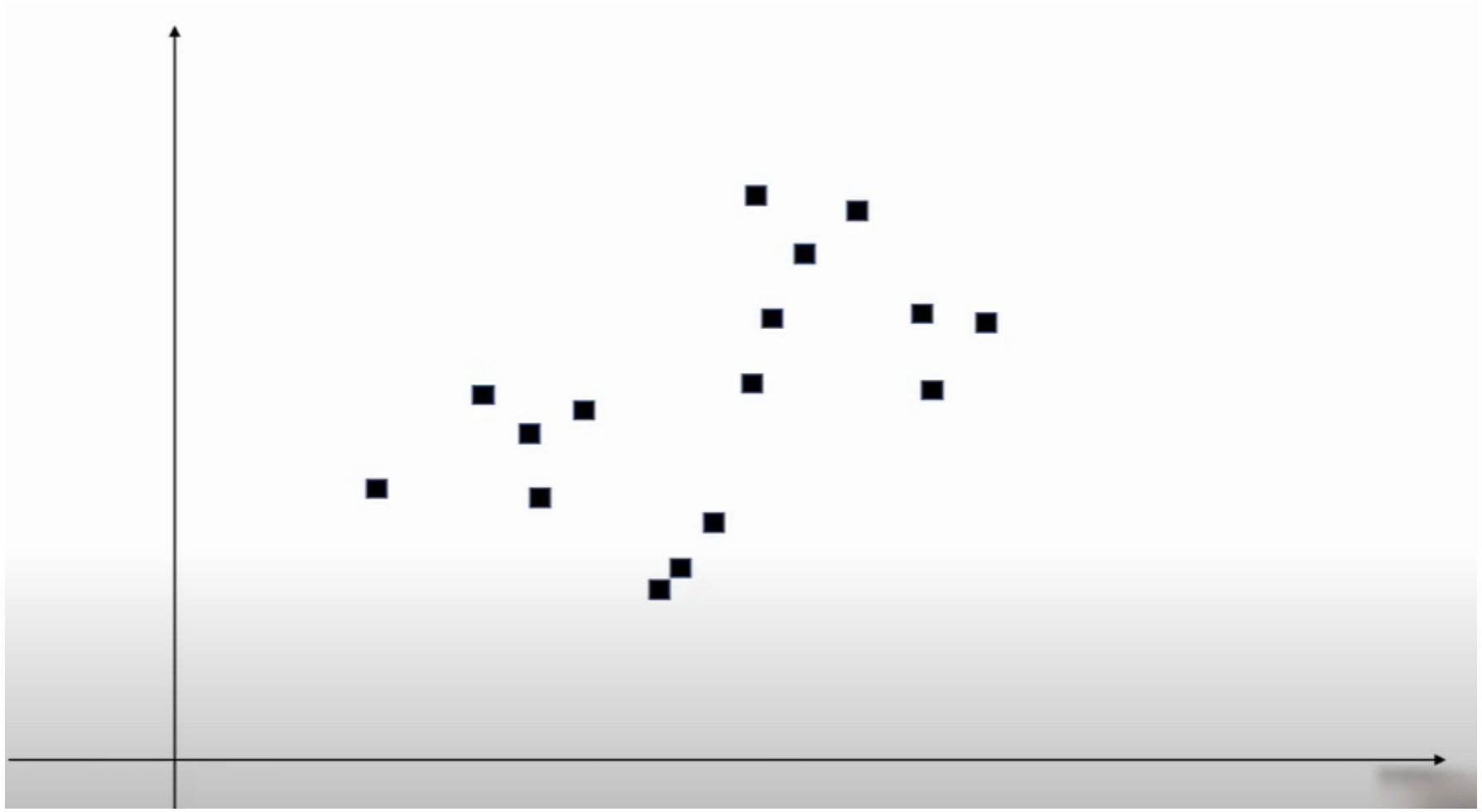
# K Means Clustering (The Intuition)



# How to determine

The correct number of clusters (k)?

# K Means Clustering (How to determine K?)



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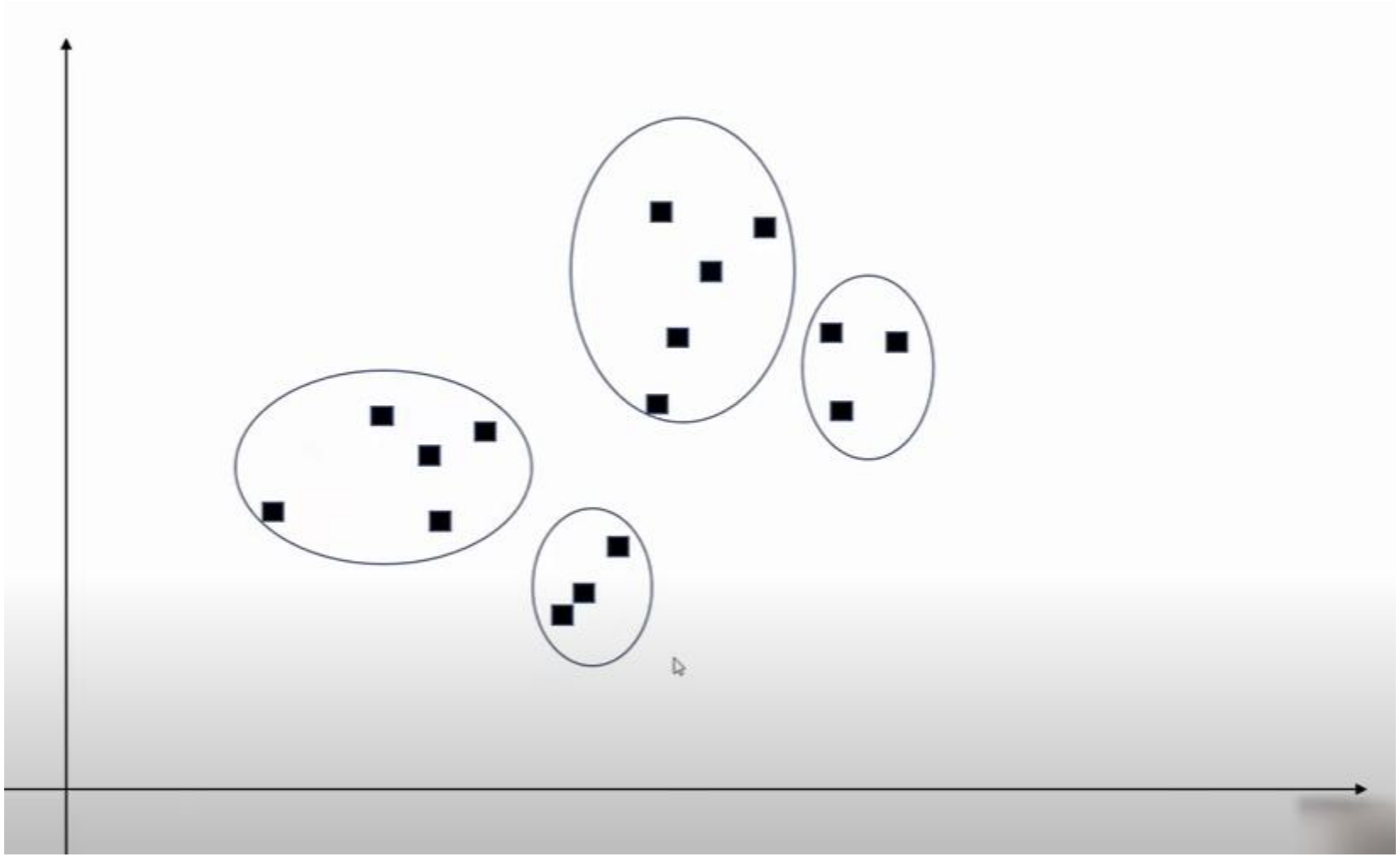

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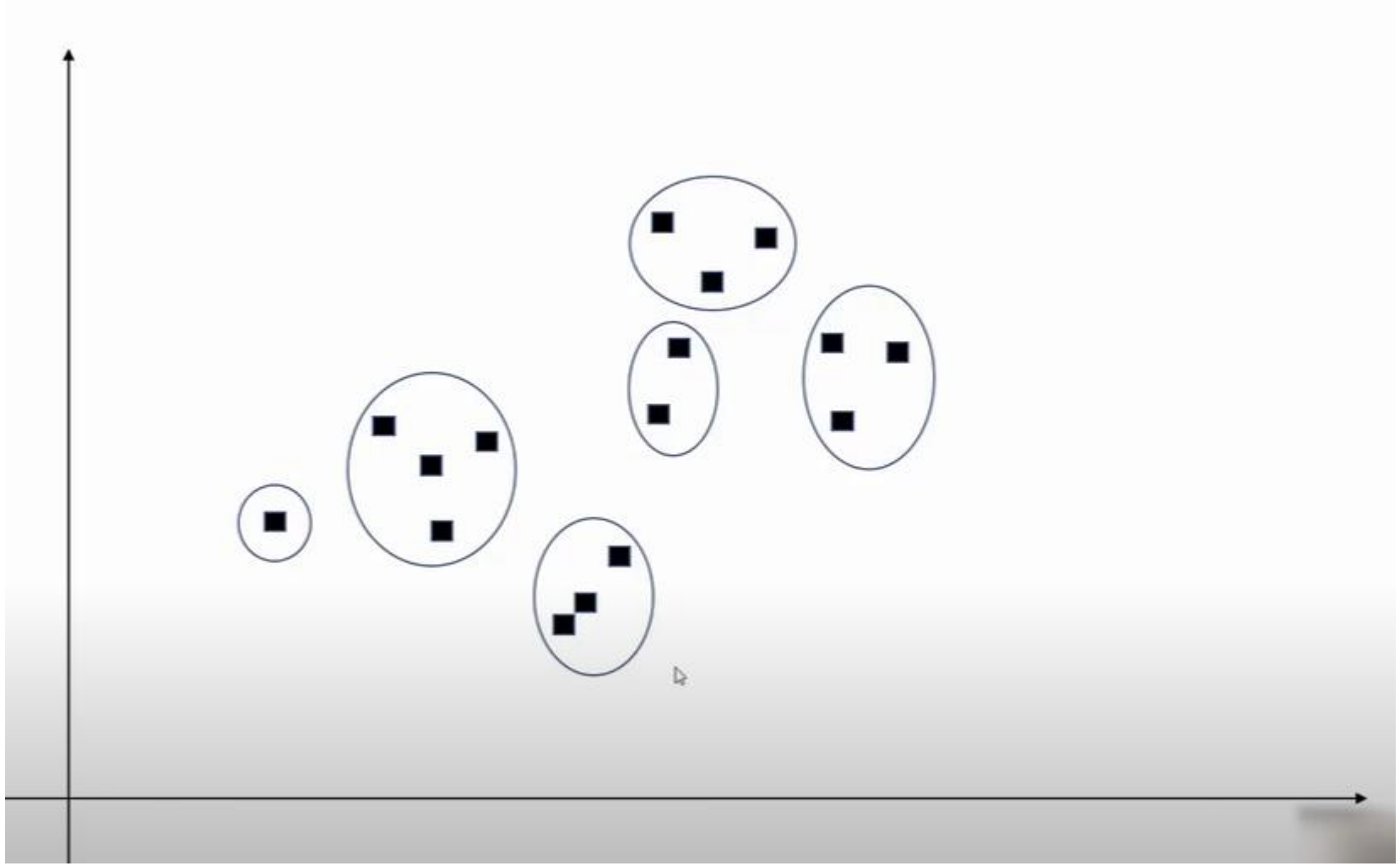

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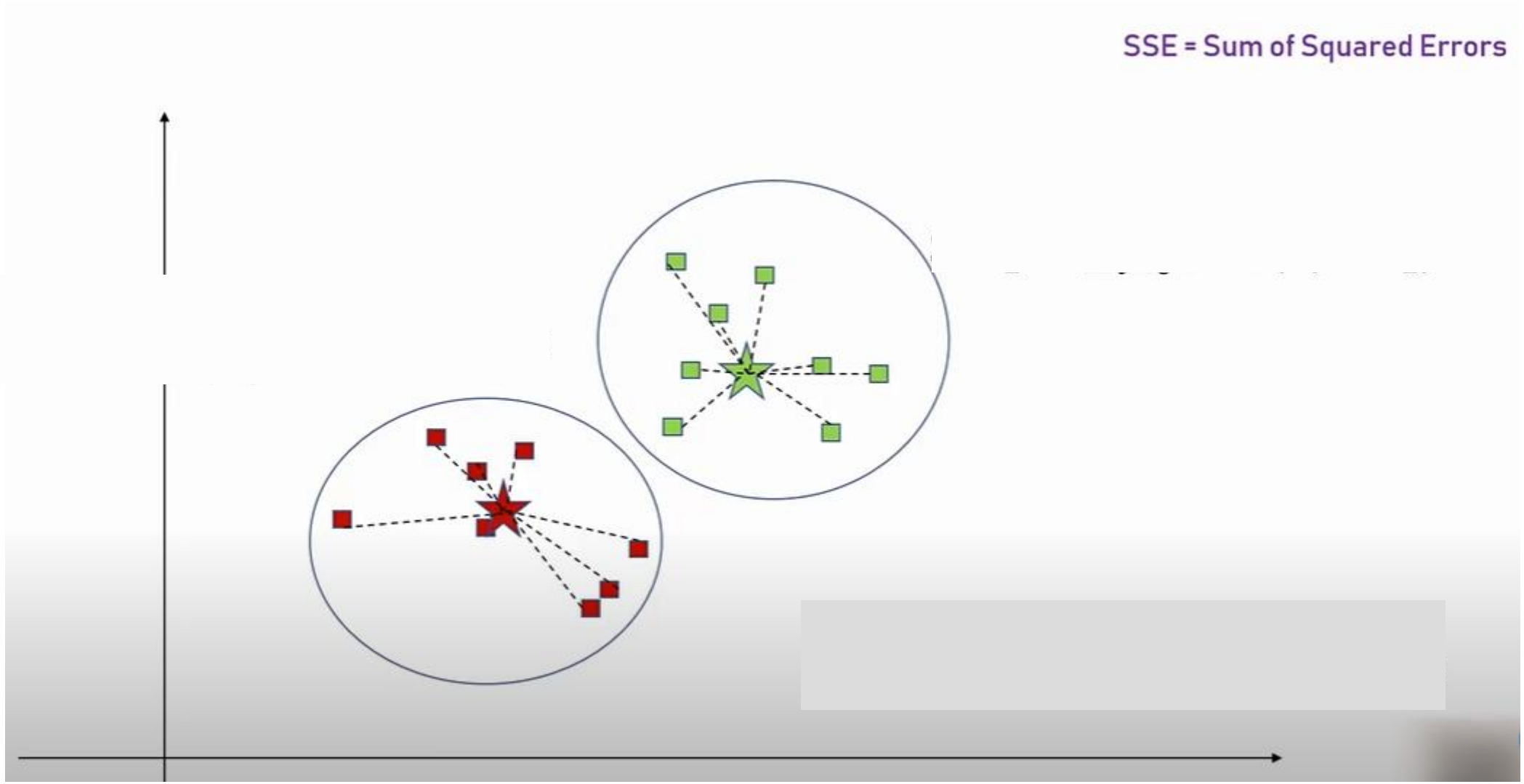

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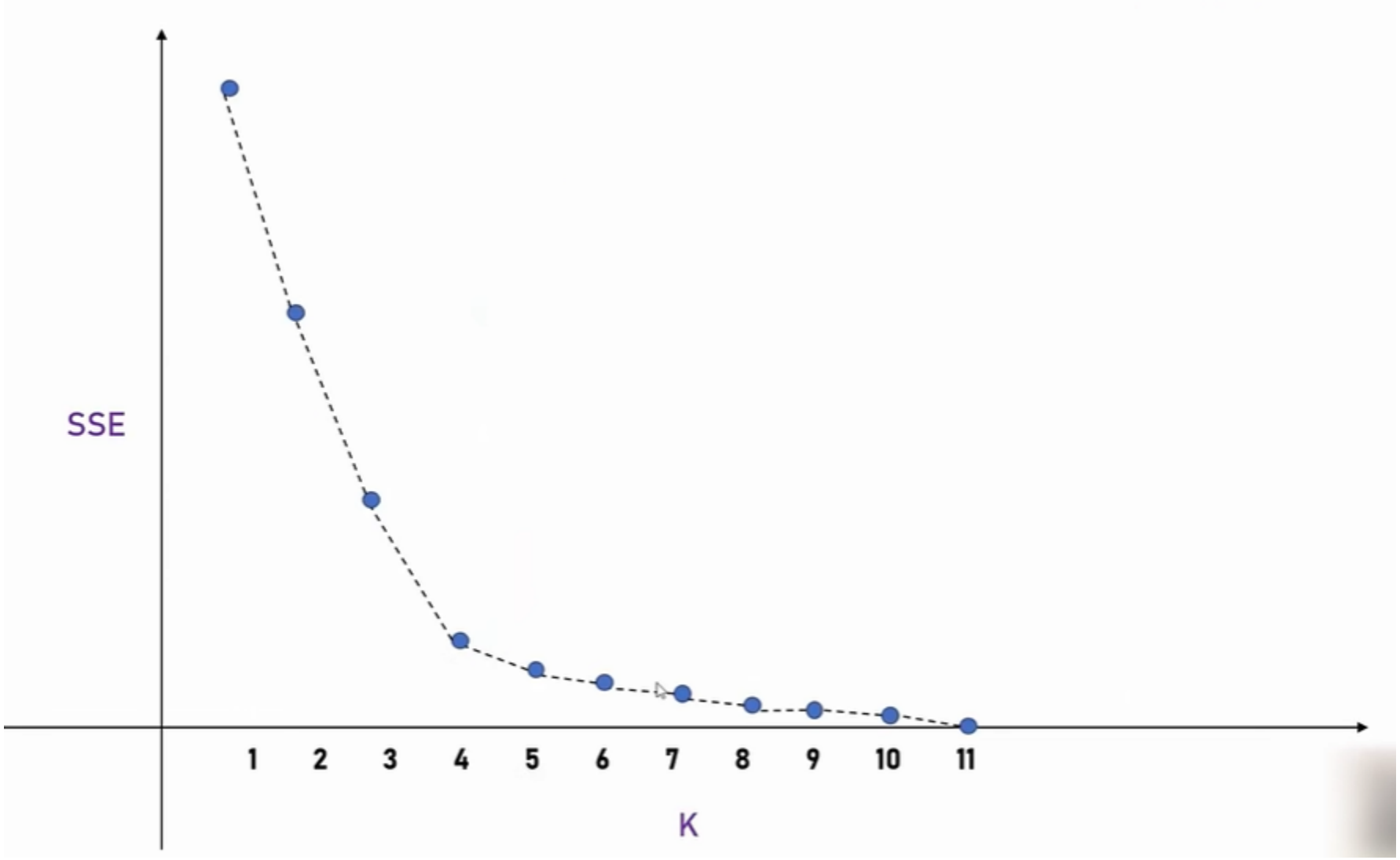

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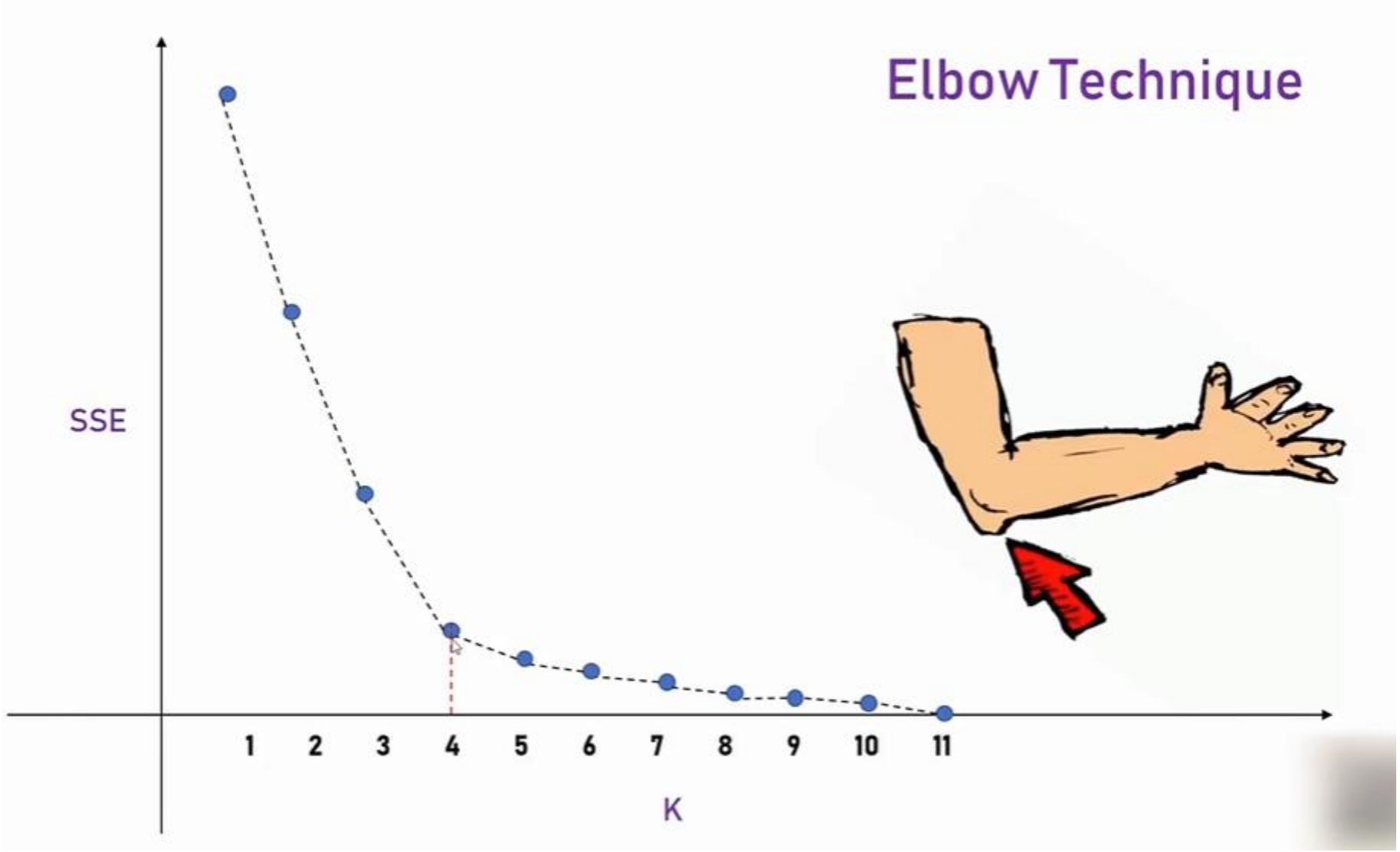

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# Hierarchical Clustering

We asked **people** how many **hours a week** they spend on **social media** platforms and in the **gym**.

| | Social media | Gym |
|---|---|---|
| Alan | 2 | 3 |
| Lisa | 5 | 2 |
| Joe | 5 | 3 |
| Max | 1 | 4 |
| Caro | 4 | 5 |

# Hierarchical Clustering (Cont..)

| | Social media | Gym |
|---|---|---|
| Alan | 2 | 3 |
| Lisa | 5 | 2 |
| Joe | 5 | 3 |
| Max | 1 | 4 |
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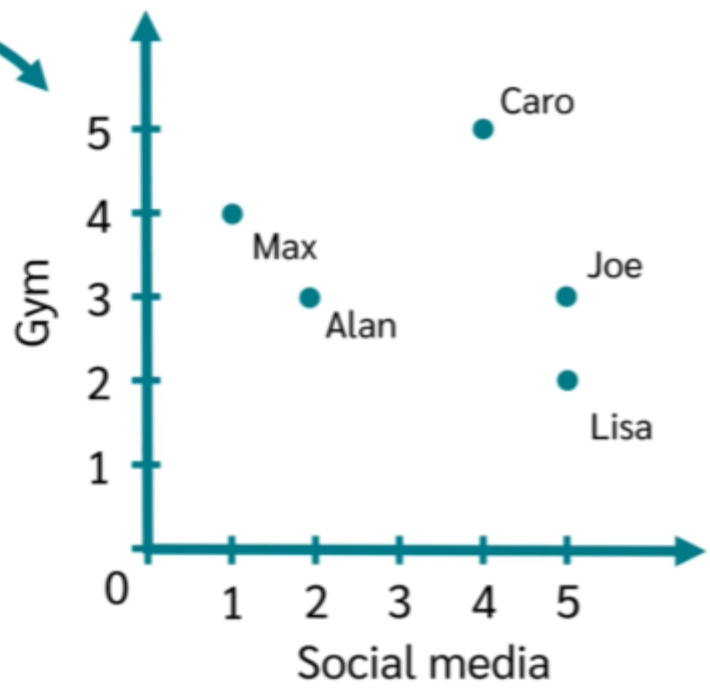
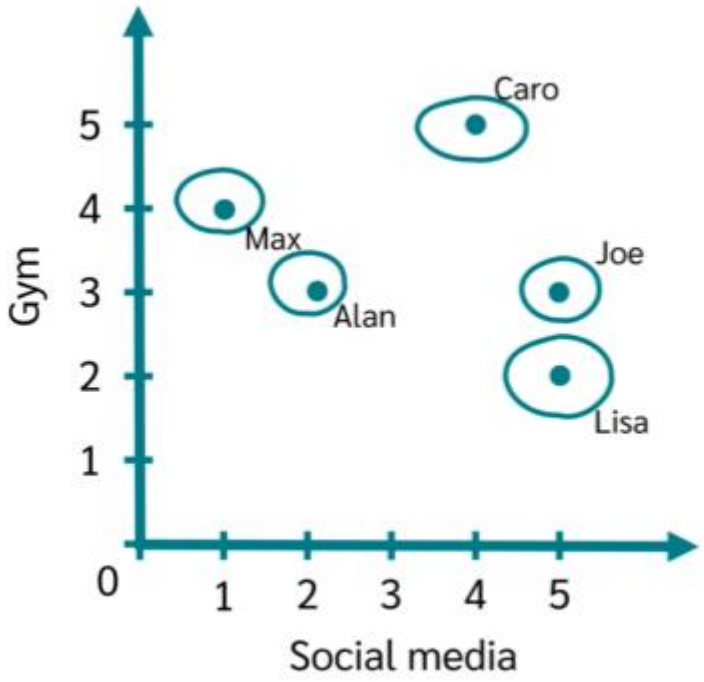


# Hierarchical Clustering (Cont..)

- First step: Assign a cluster to each individual point



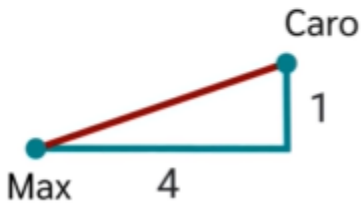

So we have as many **clusters**

# Hierarchical Clustering (Cont..)

- Measuring the distance between two points

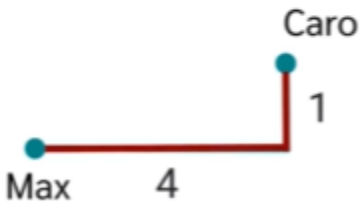
**Euclidean distance**

$$d = \sqrt{4^2 + 1^2} = 3{,}162$$



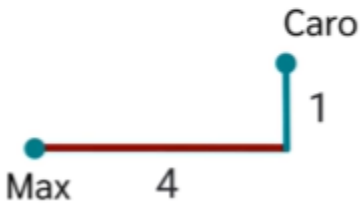

**Manhattan distance**

$$d = 4 + 1 = 5$$




**Maximum distance**

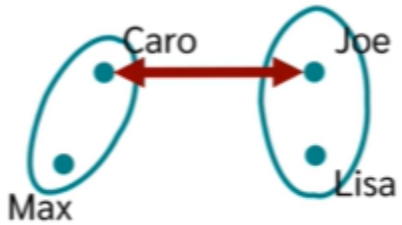
$$d = max(4\,,1) = 4$$



# Hierarchical Clustering (Cont..)

- Measuring the distance between two clusters

**Single-linkage** uses the distance between the **closest elements** in the cluster.

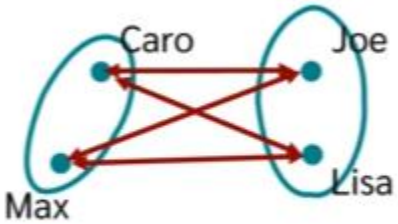



So, the distance between Caro and Joe.

**Complete Linkage** uses the distance between the **most distant elements** of the cluster.

So, between Max and Joe.

**Average linkage** uses the average of all pairwise distances.




From each combination the distance is calculated and from it the average value.

# Hierarchical Clustering (Cont..)

- The distance matrix

The distance between **Alan and Lisa** is given by:

$$d = \sqrt{(5-2)^2+(2-3)^2} = 3{,}16$$

| | Social media | Gym |
|---|---|---|
| Alan | 2 | 3 |
| Lisa | 5 | 2 |
| Joe | 5 | 3 |
| Max | 1 | 4 |
| Caro | 4 | 5 |

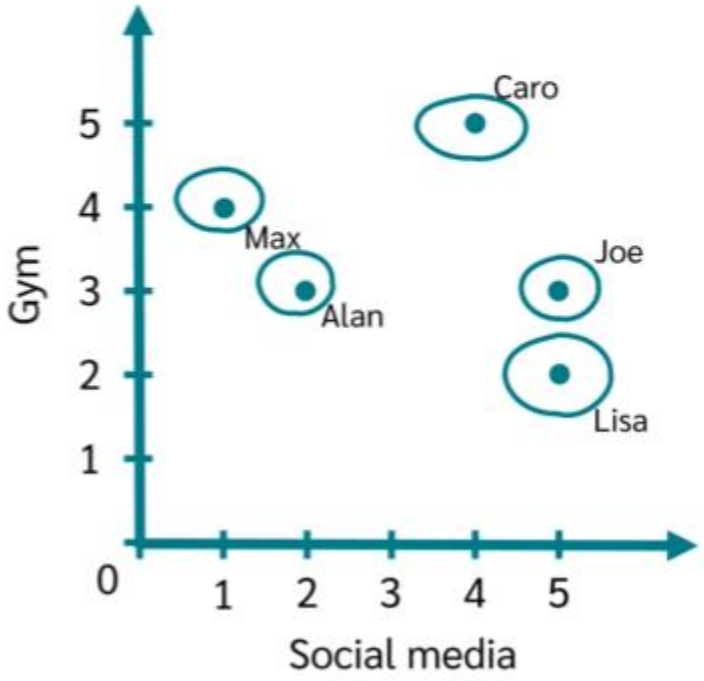
| | Alan | Lisa | Joe | Max | Caro |
|---|---|---|---|---|---|
| Alan | 0 | | | | |
| Lisa | 3,16 | 0 | | | |
| Joe | 3,00 | 1,00 | 0 | | |
| Max | 1,41 | 4,47 | 4,12 | 0 | |
| Caro | 2,83 | 3,16 | 2,24 | 3,16 | 0 |

# Hierarchical Clustering (Cont..)

Now we can **merge** the **first clusters.**

For this, we look at which two clusters have the **smallest distance** between them.

| | Alan | Lisa | Joe | Max | Caro |
|---|---|---|---|---|---|
| Alan | 0 | | | | |
| Lisa | 3,16 | 0 | | | |
| Joe | 3,00 | 1,00 | 0 | | |
| Max | 1,41 | 4,47 | 4,12 | 0 | |
| Caro | 2,83 | 3,16 | 2,24 | 3,16 | 0 |



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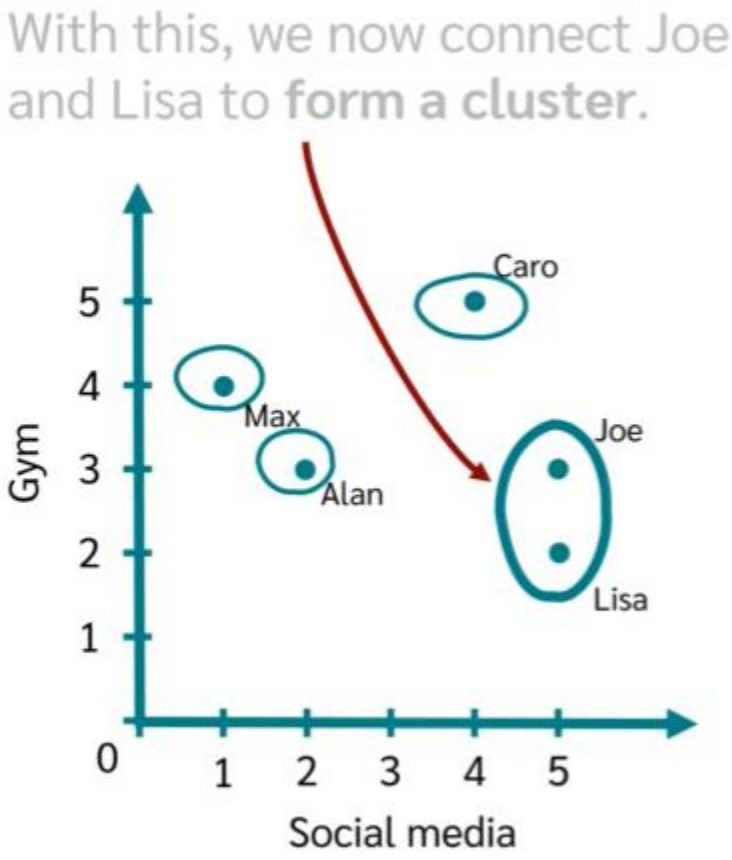
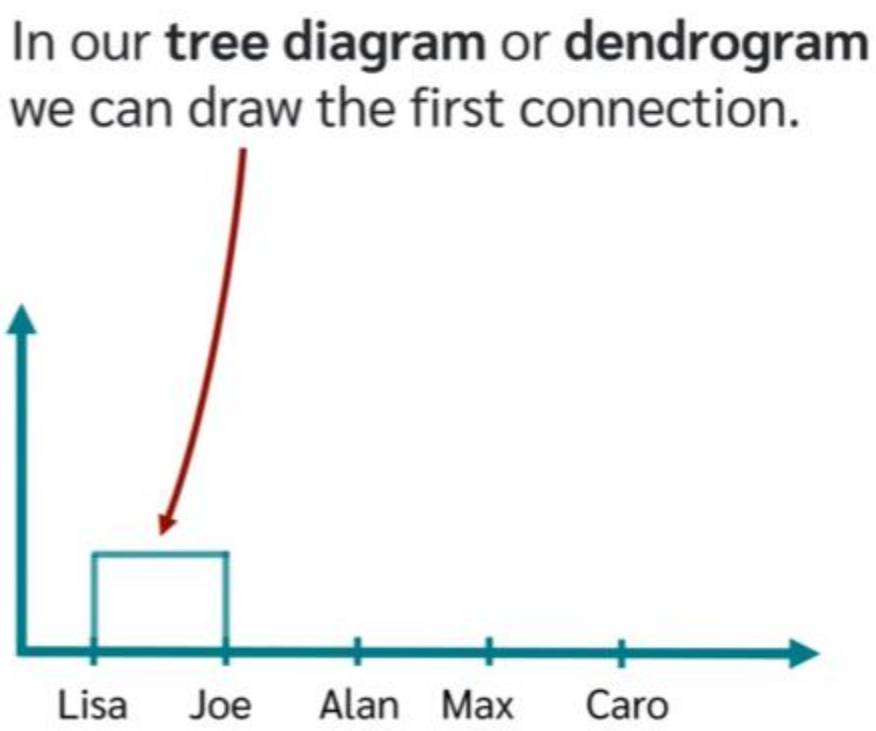
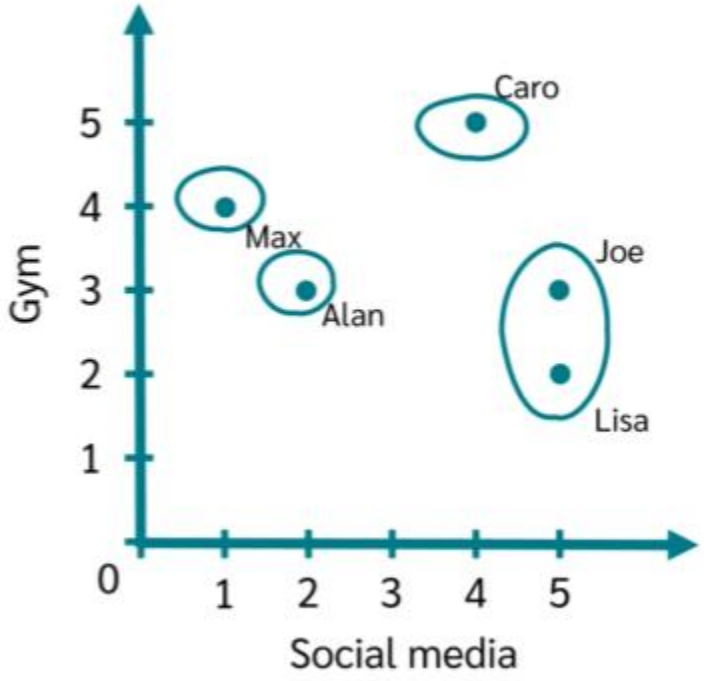


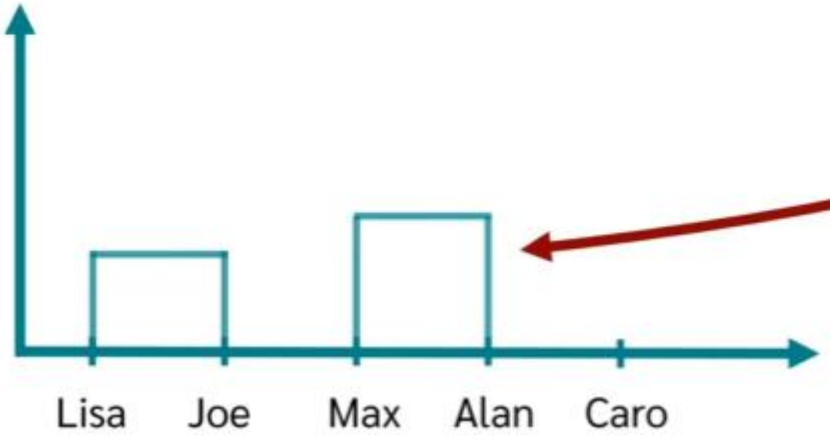
# Hierarchical Clustering (Cont..)

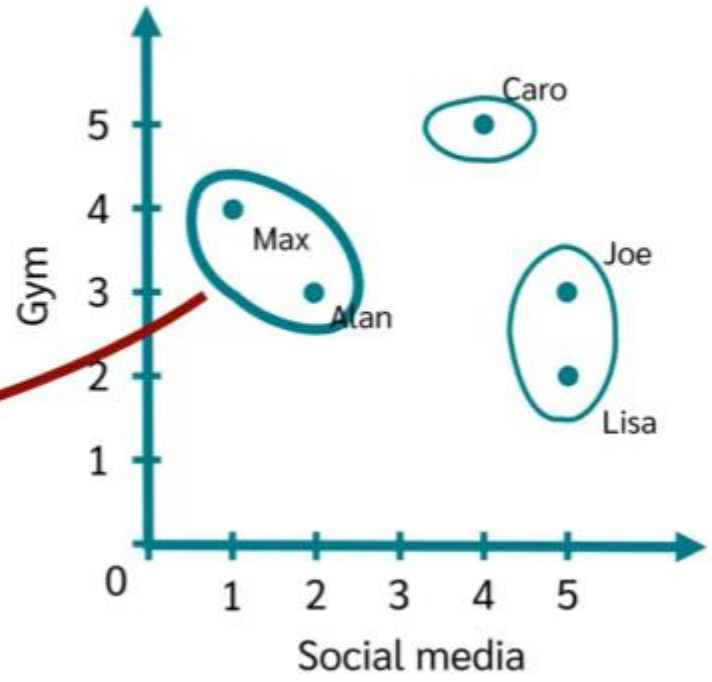
| | Alan | Lisa, Joe | Max | Caro |
|---|---|---|---|---|
| Alan | 0 | | | |
| Lisa, Joe | 3,00 | 0 | | |
| Max | 1,41 | 4,12 | 0 | |
| Caro | 2,83 | 2,24 | 3,16 | 0 |



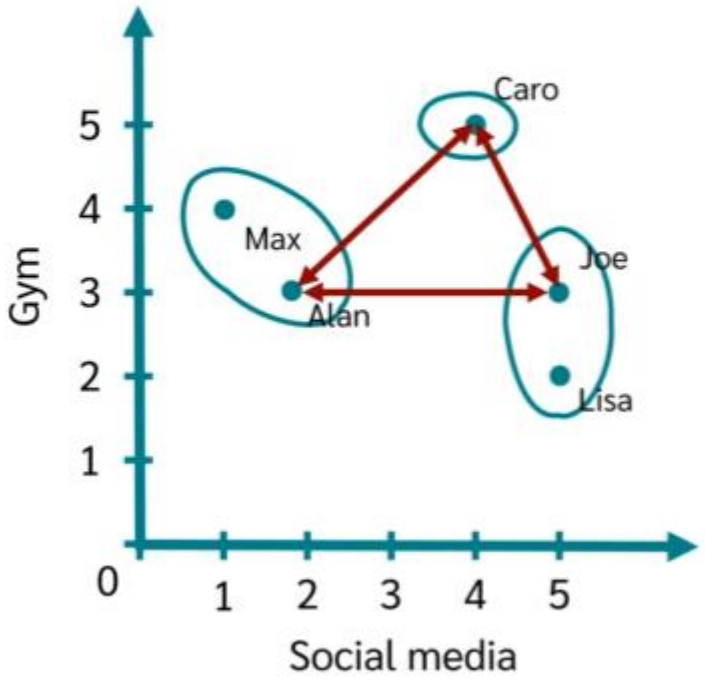
# Hierarchical Clustering (Cont..)

In our **tree diagram** or **dendrogram** we can draw the second connection.



# Hierarchical Clustering (Cont..)

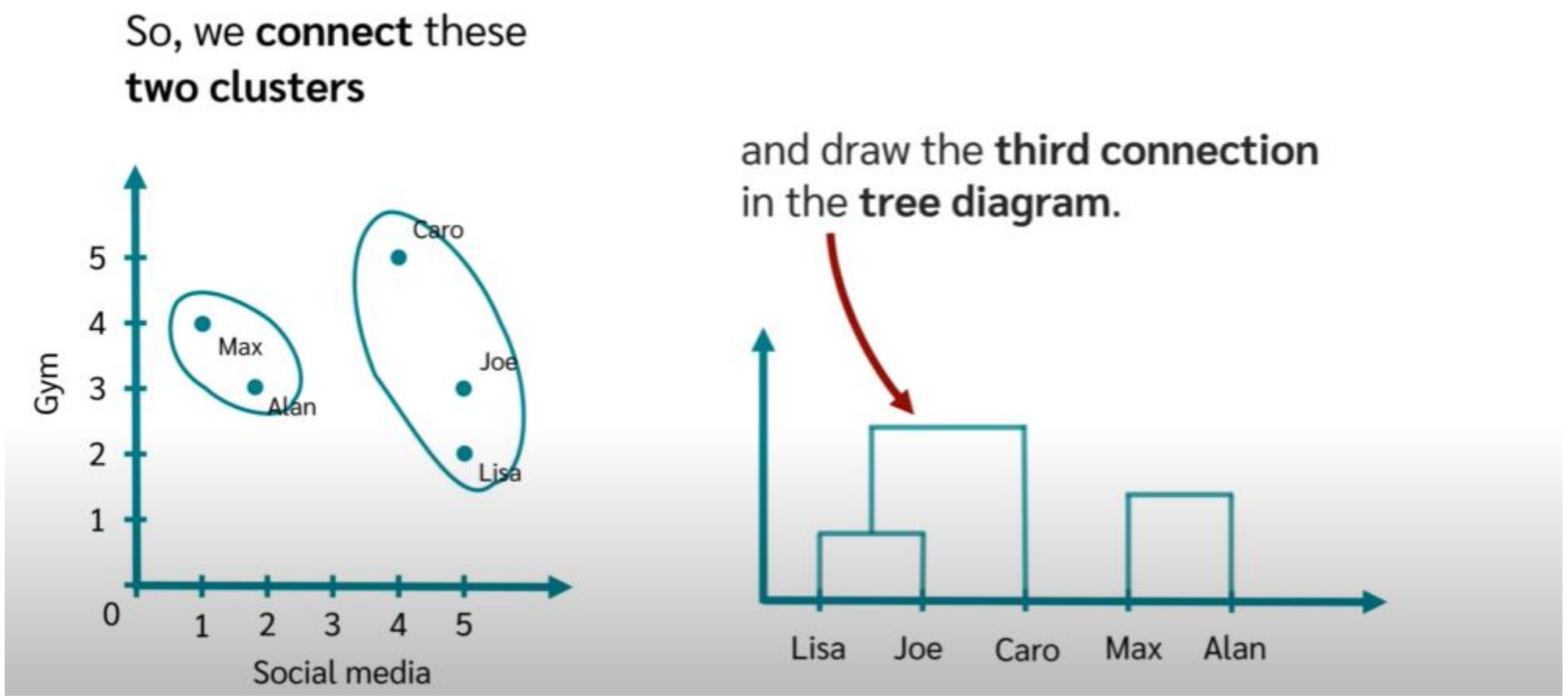



Now we **update** the **distance matrix** again.

We calculate the distance between

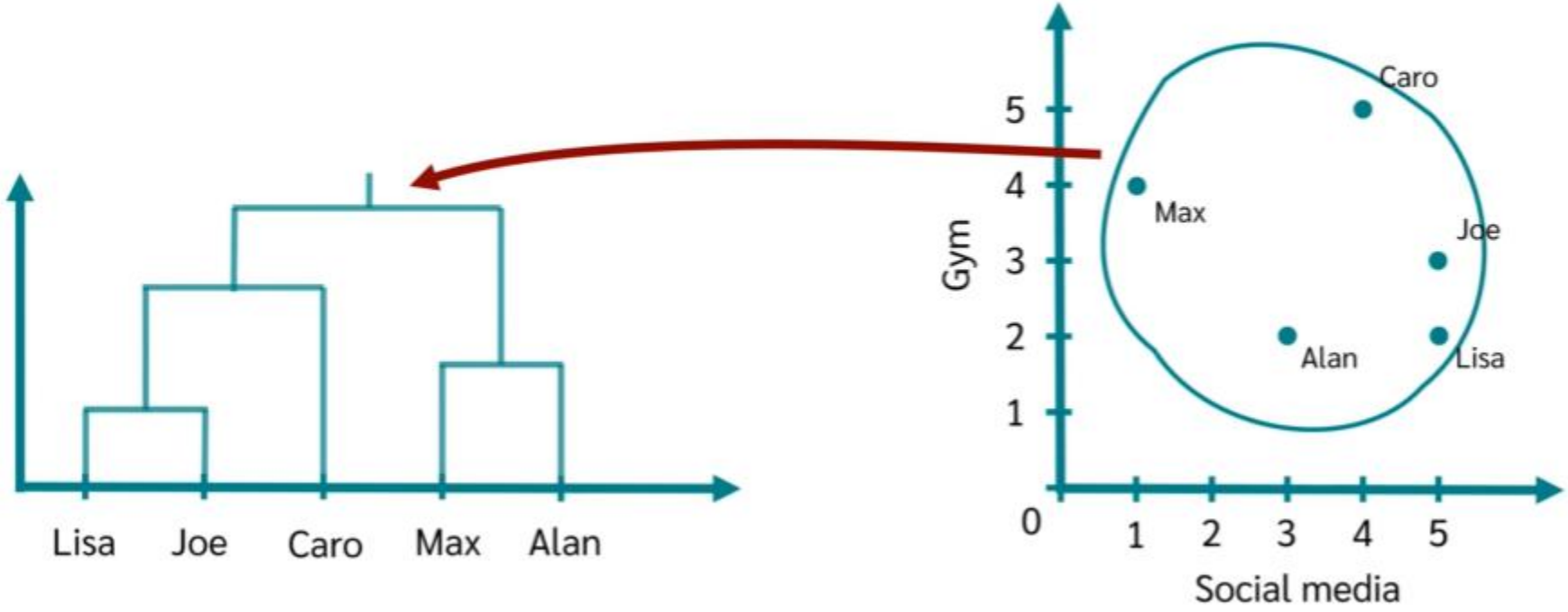
| | Lisa, Joe | Max, Alan | Caro |
|---|---|---|---|
| Lisa, Joe | 0 | | |
| Max, Alan | 3,00 | 0 | |
| Caro | 2,24 | 2,83 | 0 |

# Hierarchical Clustering (Cont..)



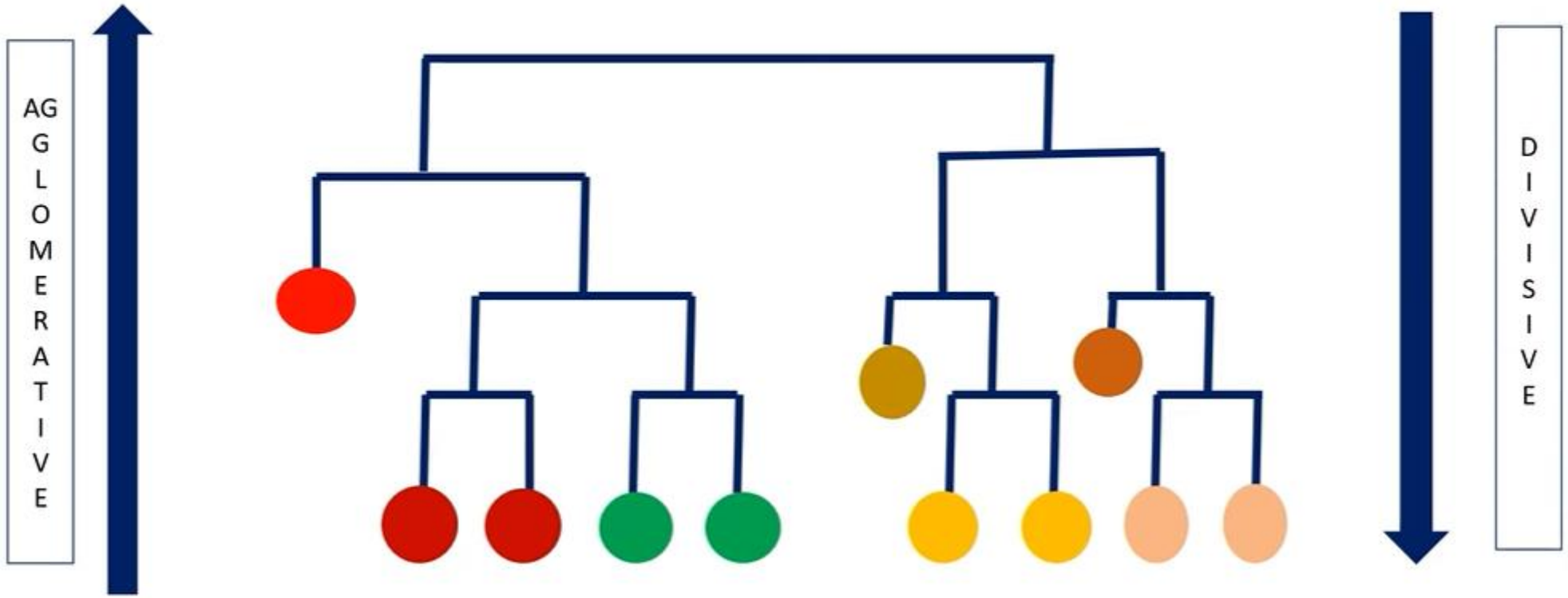
# Hierarchical Clustering (Cont..)

- Finished dendrogram

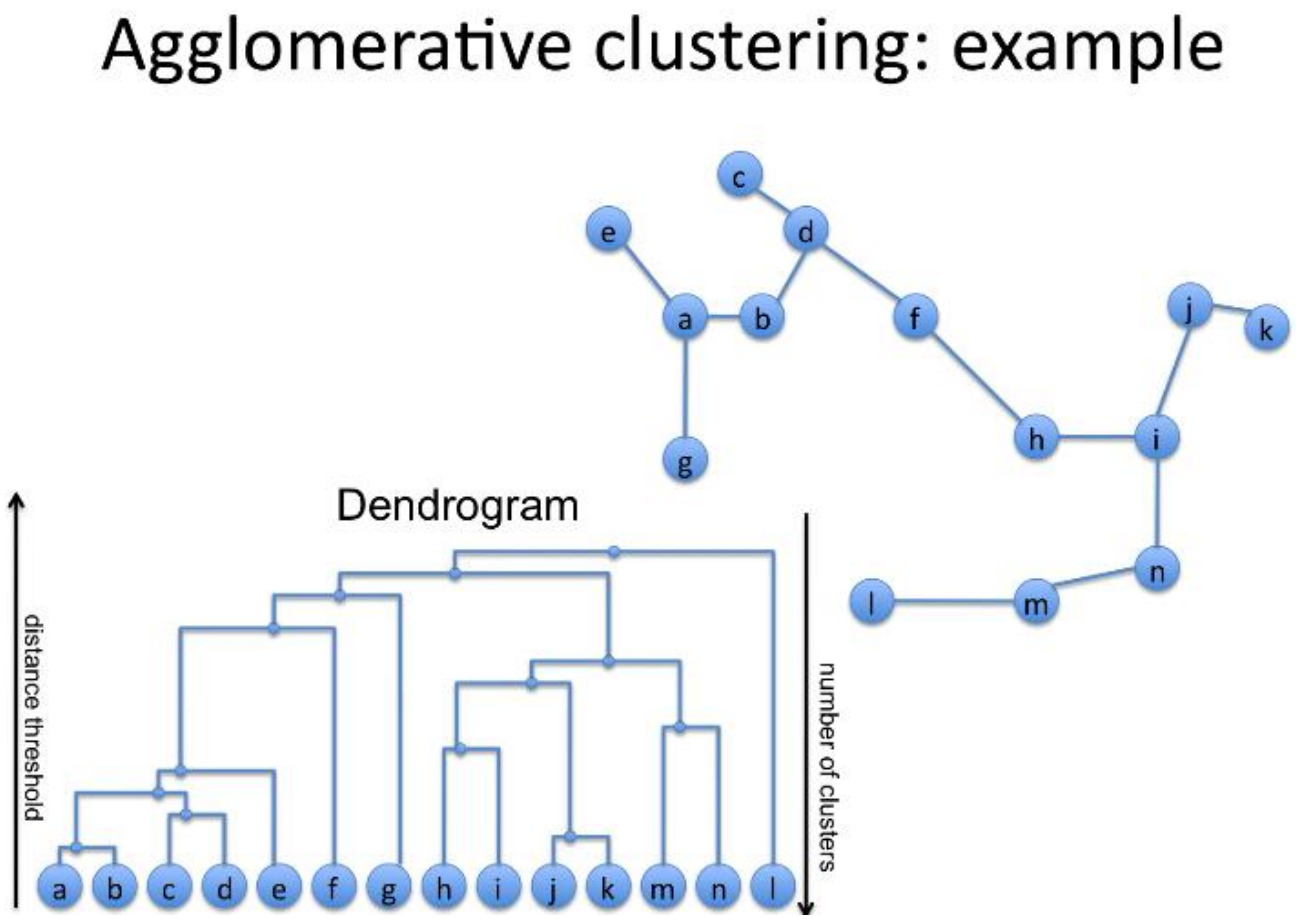


# Hierarchical Clustering (Cont..)

- Agglomerative and divisive clustering



# How to determine cluster #



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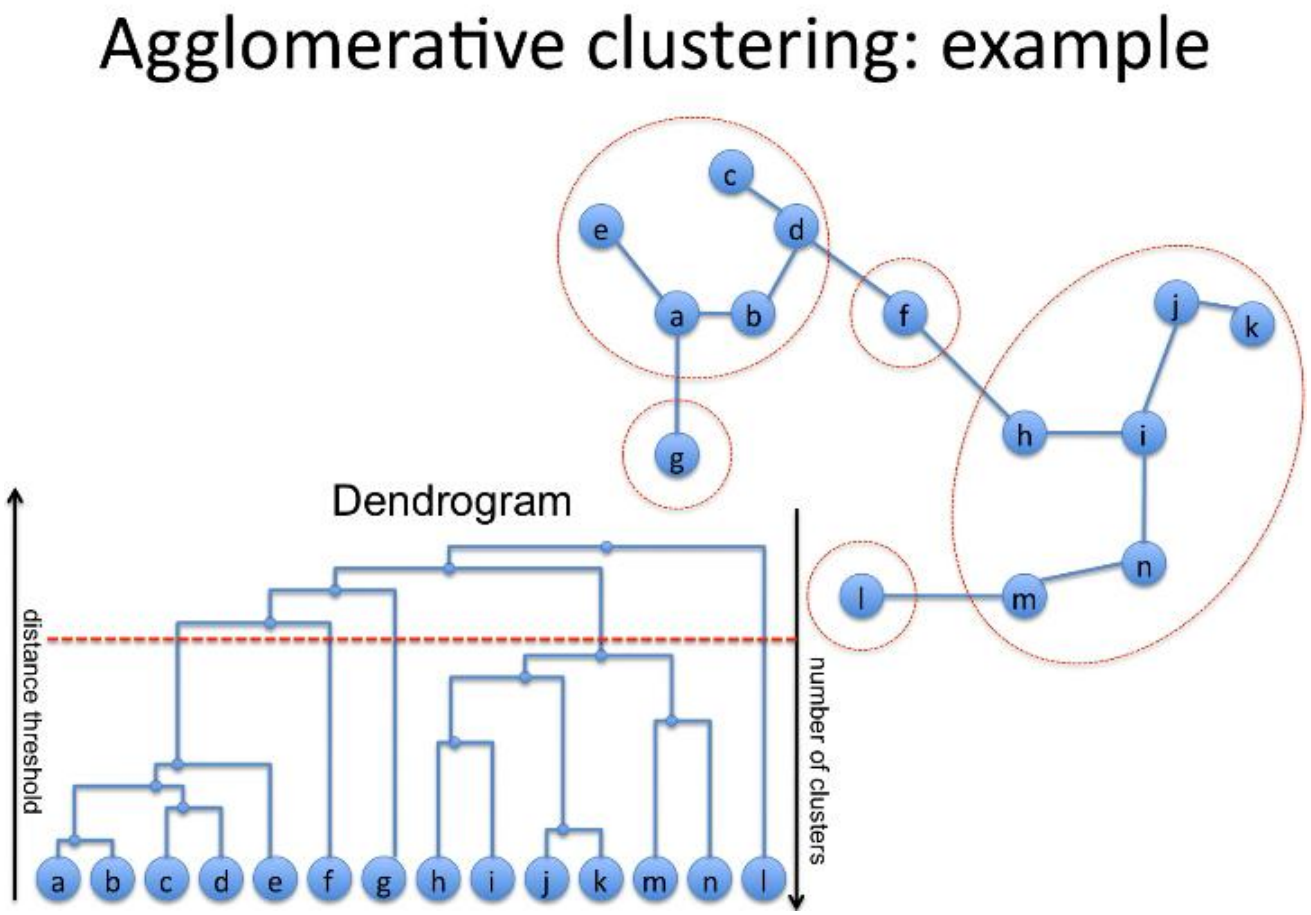
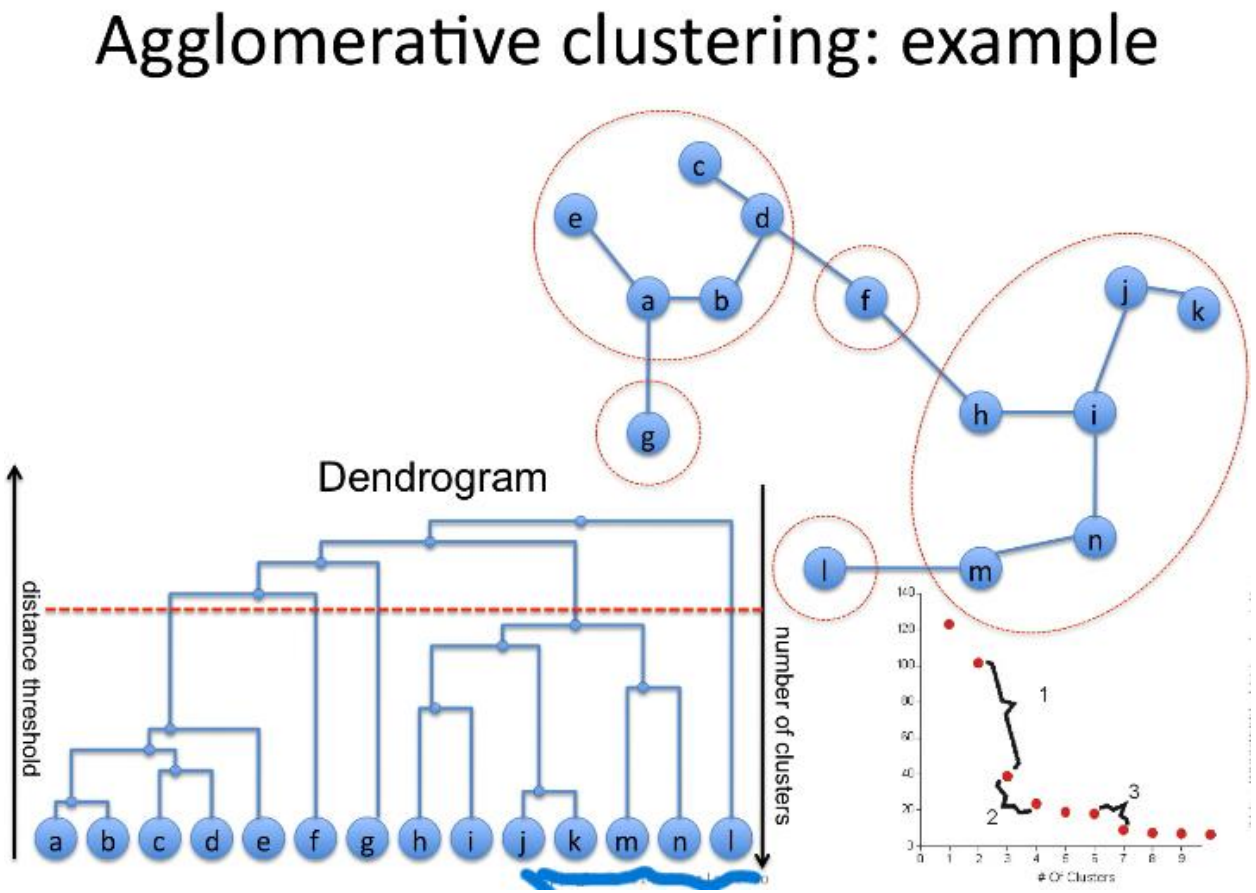
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